Topic 2.8 - Inverse Functions

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: ______________ Class: ______________ Date: ______________

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Determine whether a function is invertible on a stated domain.
- Construct and verify inverse functions algebraically, numerically, and graphically.
- Interpret inverse inputs, outputs, domains, ranges, and restrictions in context.

Section A: Foundation and representation

Question 1

Core skill | No calculator

Determine whether the relation $\{(-2,5),(0,1),(3,5),(7,-4)\}$ defines an invertible function. Justify.

Question 2

Core skill | No calculator

Find the inverse of $f(x) = 4x + 11$ and verify one composition.

Question 3

Core skill | No calculator

Find the inverse of $g(x) = \frac{x-6}{3}$.

Question 4

Core skill | No calculator

Find the inverse of $h(x) = x^3 - 8$ and state both domains.

Question 5

Core skill | No calculator

Restrict the domain of $p(x) = x^2 - 6x + 13$ to $x \geq 3$, then find $p^{-1}(x)$.

Question 6

Core skill | No calculator

Find the inverse of $r(x) = \frac{2x+1}{x-4}$ and list the excluded input of each function.

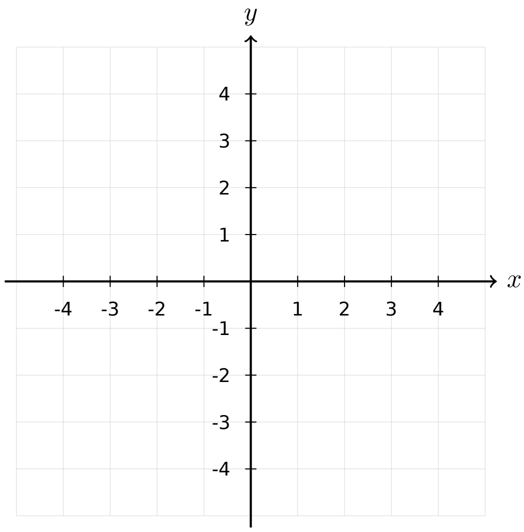
Section B: Application, comparison, and error analysis

Question 7
Core skill · No calculator

If $f(-3) = 8$, $f(1) = 0$, and $f(5) = -2$, give three values of f^{-1}.

Question 8
Graphing · No calculator

A graph of f contains points $(-1, 4), (2, 7), (6, 9)$. Plot the corresponding inverse points and describe the reflection.



Question 9
Error analysis · No calculator

A student swaps x and y in $y = 2x - 5$ and stops at $x = 2y - 5$. Explain why this is incomplete.

Question 10
Context | No calculator

A taxi model $C(d) = 8 + 2.4d$ gives cost in QAR from distance d in km. Find C^{-1} and interpret $C^{-1}(32)$.

Question 11
Core skill | No calculator

Let $f(x) = \sqrt{x-1} + 3$. Find the inverse and verify the domain-range swap.

Question 12
Parameter reasoning | No calculator

For what values of a is $f(x) = ax + 4$ invertible on $\mathbb{R}$? Give the inverse.

Section C: AP-style multiple choice

MCQ 1
No calculator

Which function is not one-to-one on all real numbers?

A. $3x - 1$

B. $x^3 + 2$

C. 2^x

D. $(x-4)^2$

MCQ 2

No calculator

If $f(6) = -3$, which statement is true?

A. $f^{-1}(6) = -3$

B. $f^{-1}(-3) = 6$

C. $f(-3) = 6$

D. $f^{-1}(3) = -6$

MCQ 3

No calculator

On $x \geq 0$, the inverse of $f(x) = x^2 + 5$ is:

A. $\sqrt{x} + 5$

B. $\sqrt{x - 5}$

C. $-\sqrt{x - 5}$

D. $x^2 - 5$

MCQ 4

No calculator

The graphs of a function and its inverse are symmetric about:

A. The x-axis

B. The y-axis

C. $y = x$

D. $y = -x$

Section D: Free-response reasoning

FRQ 1 - Restricted quadratic inverse

No calculator | 6 points

Let $f(x) = (x+2)^2 - 7$ with domain $x \geq -2$. (a) Explain why the domain restriction is needed. (b) Find $f^{-1}(x)$ and its domain. (c) Verify $f^{-1}(f(x)) = x$ on the stated domain. (d) State the reflection relationship between the two graphs.

FRQ 2 - Rational inverse and context

No calculator | 6 points

A calibration function is $R(s) = \dfrac{5s+12}{s+2}$ for $s \neq -2$. (a) Find R^{-1}. (b) State the excluded input and output values of R. (c) Verify one composition algebraically. (d) Explain what $R^{-1}(4)$ means if s is a sensor setting and R is the displayed reading.